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Reducing Water in Gas Wells with Autonomous Inflow Control Technology

T. T. Larsen, InflowControl AS, Porsgrunn, Norway; K. Langaas, Aker BP ASA, Norway; T. C. Dhital and I. M. Ismael, InflowControl AS, Porsgrunn, Norway

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Abstract

Energy is critical for the global population, and natural gas is a reliable, affordable and transportable source. Water breakthroughs in gas wells can reduce productivity and recovery, leading to early well shut in. A new Autonomous Inflow Control Technology for gas wells (gas AICT) has been developed to autonomously detect and choke unwanted water production while allowing gas and condensate production, improving both economic and sustainability measures of gas wells.

This novel AICT leverages the Hagen-Poiseuille equation and Bernoulli's principle to differentiate between gas, condensate, and unwanted water. The gas AICT autonomously chokes zones with water production. Extensive flow performance tests have been conducted as part of the qualification program, initially in a multiphase flow loop with model fluids and subsequently with hydrocarbon gas, condensate, and formation water under downhole conditions. A reservoir and well simulation model, based on a Norwegian gas discovery, was constructed to evaluate the impact of the gas AICT. This model also aids in understanding future development directions for the technology.

The experimental results demonstrate that the gas AICT effectively chokes water while remaining open for gas and partially open for light condensate. The valve exhibits an acceptable flow rate for gas, which can potentially be further enhanced in the future. Reservoir simulation modelling predicts the performance of the gas AICT under downhole conditions and explores various scenarios for the lower completion. For the long horizontal gas wells in this study, the impact of annulus phase segregation is quantified. Potential solutions for the clean-up of heavier completion fluids are also briefly discussed.

To our knowledge, this is the first published flow performance test of an AICT that autonomously detects and chokes water while allowing gas and light condensate at downhole conditions. This technology has the potential to extend well life and increase gas recovery, with estimates in the range 13-37% in the field case studied. Additionally, water production is significantly reduced. The reduction in water production and the improvement in gas recovery (reducing the need for new wells by enhancing recovery from existing ones) provide clear environmental benefits. In summary, the gas AICT studied here shows clear potential to enhance both economic and sustainability outcomes for new gas wells at risk of water production.

Introduction

Autonomous inflow control technology (AICT), which operates without the need for electronics or other means of communication or control, has demonstrated its market value by efficiently choking back unwanted fluids in oil wells. Currently, there is no proven autonomous technology for gas wells with potential water problems, although it has been discussed as a potential application for density-based AICT (Killie and Grüner, 2024). Using the same principles as the proven AICT concept for oil wells, this paper presents the development and testing of a novel AICT for gas wells. Hereafter, we will refer to this new technology as the "gas AICT" or simply "the valve".

The valve's autonomy is regulated by a secondary flow path, known as the pilot flow, which runs parallel to the main flow within the valve. The pilot flow comprises two constant flow resistances in series: the turbulent flow element (TFE) and the laminar flow element (LFE), as illustrated in Fig. 1. Depending on the viscosity and density of the incoming fluid, the pilot branch generates an intermediate pressure level between the two flow elements (P2), which is used to control the choking level in the valve. This effect is entirely based on fluid mechanical principles and is fully reversible. The valve can be adjusted to match the specific fluid conditions in a well.

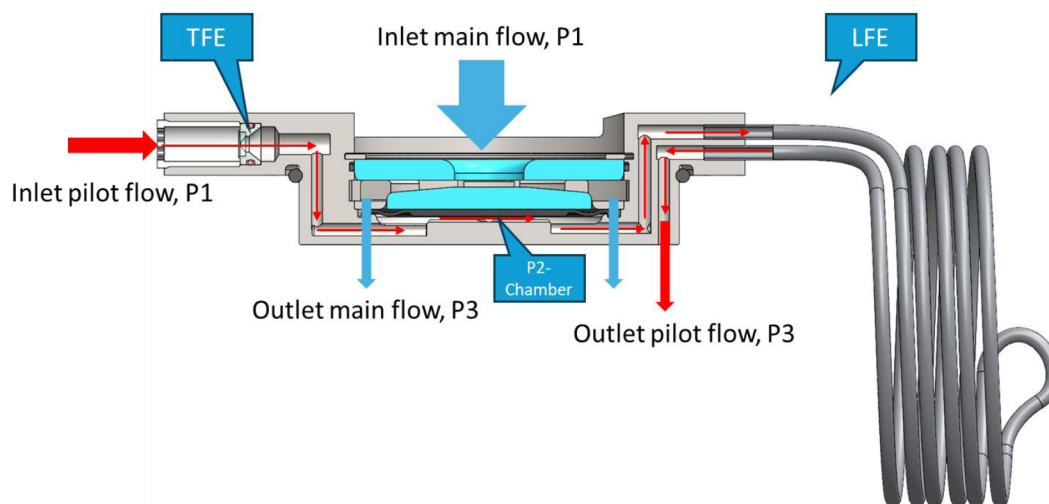


Figure 1—Section view of the AICT for gas wells showing the main flow (blue arrows) and pilot flow path (red arrows).

For a more detailed description of the similar technology for oil wells, see Aakre et al. (2014) or Elverhøy et al. (2018). Fig. 2 below illustrates the flow paths of the valve in a simplified process flow diagram.

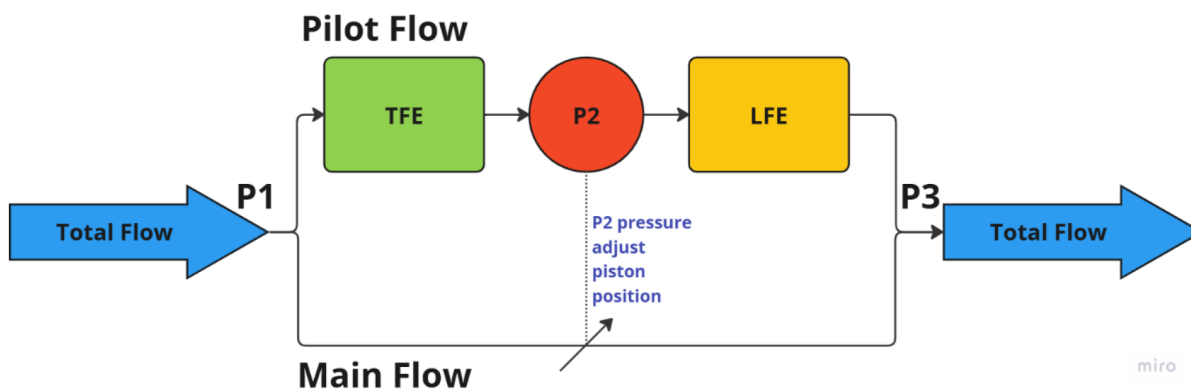


Figure 2—Process flow diagram of the flow path in the AICT showing the intermediate pressure P2 that controls the level of choking.

Test rigs and conditions

Inhouse test loop with model fluids

Initial valve development utilized a dedicated test loop with model fluids such as water and compressed air. The facility is designed to closely replicate downhole fluid conditions by precisely controlling pressure and temperature. Configured for both single- and multiphase testing, the open-loop rig includes fluid mixing prior to the test section, which simulates a screen joint to capture realistic inflow conditions. Advanced instrumentation—Coriolis flow meters, temperature and pressure sensors, and high-accuracy differential pressure transmitters—ensures comprehensive monitoring of all relevant parameters.



Figure 3—Inhouse test loop for model fluids.

Table 1—Test summary during development phase.

Test	Description
Gas (compressed dry air)	0 – 20 bar differential pressure at different absolute pressures (80-170 bara)
Water	0 – 20 bar differential pressure at different temperatures (20-90°C)
Multi-phase (gas-water)	At discrete differential pressures up to 15 bar.

The fluid properties of the model fluids were compared to the density and viscosity of the actual fluids utilized in the multi-phase flow loop tests, as shown in Fig. 4. The results indicate that model fluids can be adjusted by varying pressure and temperature to closely replicate the properties of real fluids.

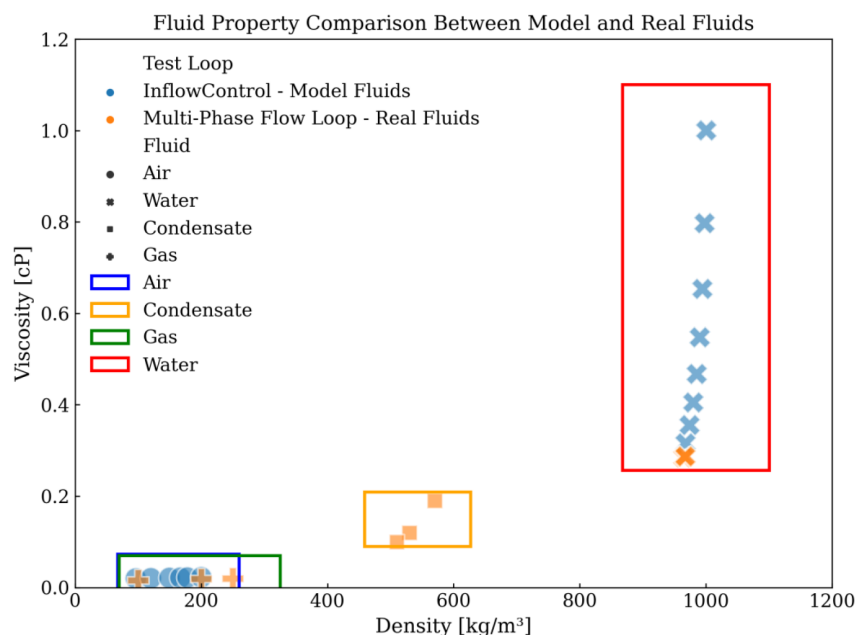


Figure 4—Plot showing a comparison of viscosity and density of model fluids versus real fluids.

Multi-phase flow loop with live hydrocarbons

The multi-phase flow is a closed loop system consisting of a multi-phase pump as the driving unit, a heat exchanger, a three-phase separator (water, condensate and gas) and a test vessel where the test valve is placed in a screen joint. The loop is equipped with flow meters for each of the phases together with temperature, differential pressure and pressure sensors around the test unit. The test facility is designed to test AICT at full scale and at downhole conditions, see [Lauritzen and Martiniussen \(2011\)](#) for further description.

Table 2—Test matrix single-phase tests

Phase	Differential pressure [bar]
Gas	0.5 – 40
Condensate	0.5 – 40
Water	0.5 – 40

Table 3—Fluid properties in the multi-phase flow loop test. Averaged values.

	Gas			Condensate			Water		
	125 bar	180 bar	200 bar	125 bar	180 bar	200 bar	125 bar	180 bar	200 bar
Density [kg/m ³]	100	200	250	570	530	510	993	990	990
Viscosity [cP]	0.016	0.019	0.020	0.19	0.12	0.10	0.30	0.30	0.30
Temperature [°C]	100	100	100	100	100	100	100	100	100

Test results

In [Fig. 5](#) below, a verification of the flow measurements is shown using a nozzle with 3 mm diameter and a theoretical discharge coefficient (C_d) of 0.88. All phases would fit better a slightly lower C_d value, and we know there is some uncertainty around the theoretical value. Condensate is a new fluid experience for

the rig, and volumetric rates are seen to deviate more than water at higher differential pressures. The actual measured points are in good agreement at lower differential pressures for gas but shows some deviations at higher differential pressures, most likely explained by an increase in gas density (constant gas density assumed in match here). No corrections have been done to the gas AICT results based on this benchmark test.

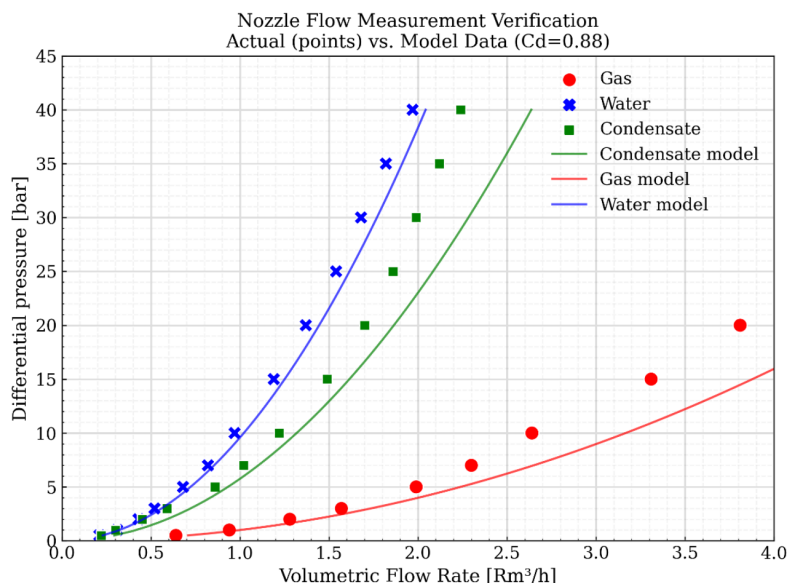


Figure 5—Verification of flow measurements in the multi-phase flow loop using a known sized nozzle/orifice of $d=3.0$ mm with a $C_d=0.88$. Points are data points while solid lines are model.

Fig. 6 presents the experimental results for the gas AICT using real fluids under actual conditions. The results show that the valve stays closed to water and partially open to condensate. The measured water origins from the pilot branch. The pilot branch always remains open to enable full reversibility and autonomy.

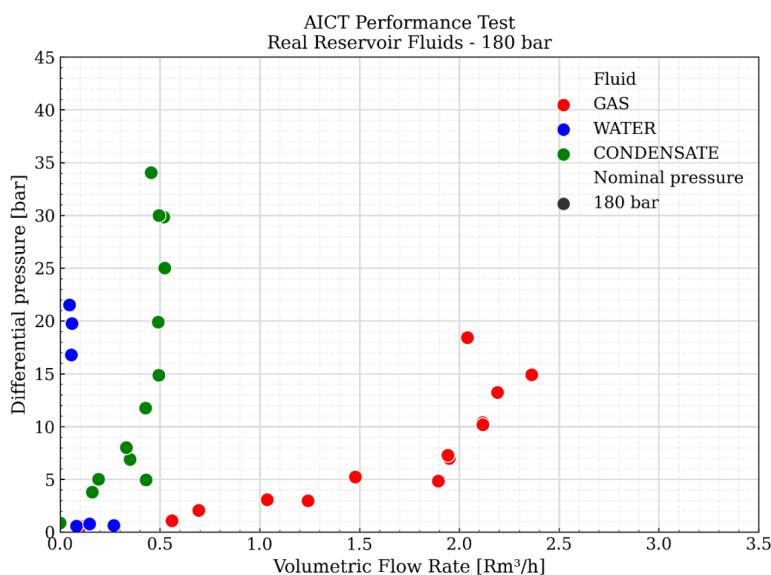


Figure 6—Performance plot at real reservoir conditions with real hydrocarbons at 180 bar. The performance plot shows how the valve is open for gas, partially open for condensate and closed for water (pilot line flow only).

Fig. 6 illustrates how well the valve responds to change in fluid properties. One of the limits that were evident during testing was the efficiency of the three-phase separator at high pressure and high flow

rates. Especially at 200 bar pressure, there is clear evidence that there was high liquid carry-over from the separator. This can be seen on the density measured on the gas line in Fig. 7 while pressure was kept constant. The valve is at this condition essentially operating in the two-phase region with a fair amount of condensate mixed in the gas phase.

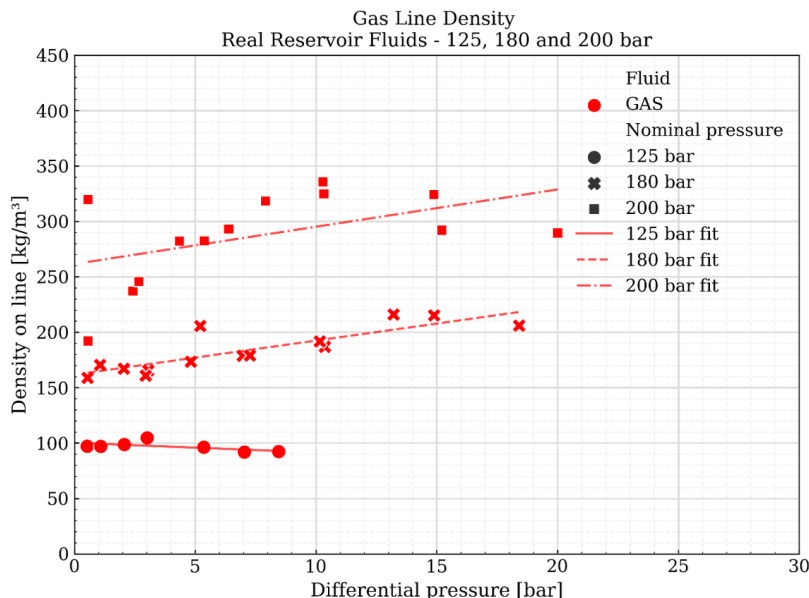


Figure 7—Measured density of gas. At the higher pressures (180 and 200 bar), the gas entrains some condensate from the separator into the gas line, observed as increased gas density at larger differential pressure (i.e. higher velocities) over the valve.

It is observed that, despite significant differences in gas line density, the mass throughput of the valve remains similar across all pressure levels. Fig. 8 below shows the volumetric flow rate at standard conditions (15°C, 1 atm), assuming a constant molar mass of the gas mix. Such consistent valve behaviour at different pressures ease the implementation of the valve into reservoir simulators.

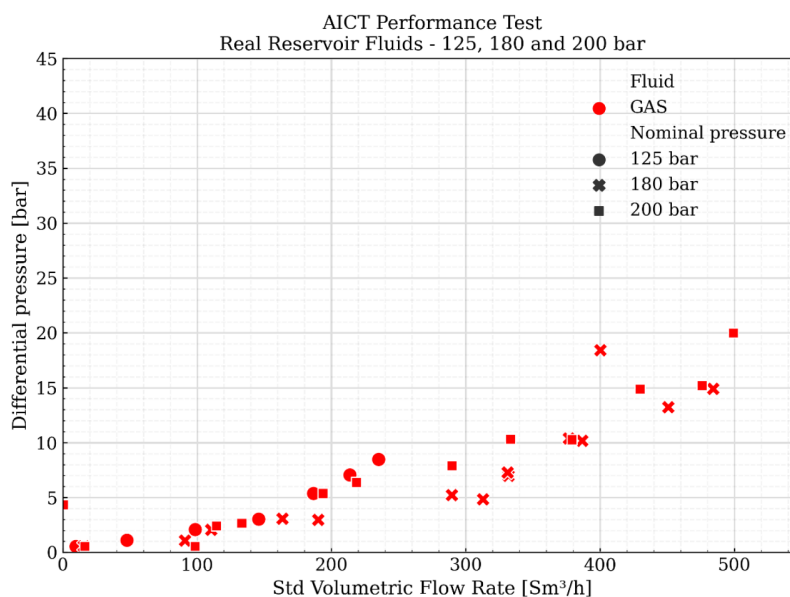


Figure 8—Performance plot of the valve showing flow rate at standard conditions. The valve maintains consistent and predictable performance within this pressure range.

Comparison between model fluids and real fluids

Fig. 9 provides a direct comparison of valve performance under test conditions using compressed air at 170 bar and with actual reservoir gas at 180 bar.

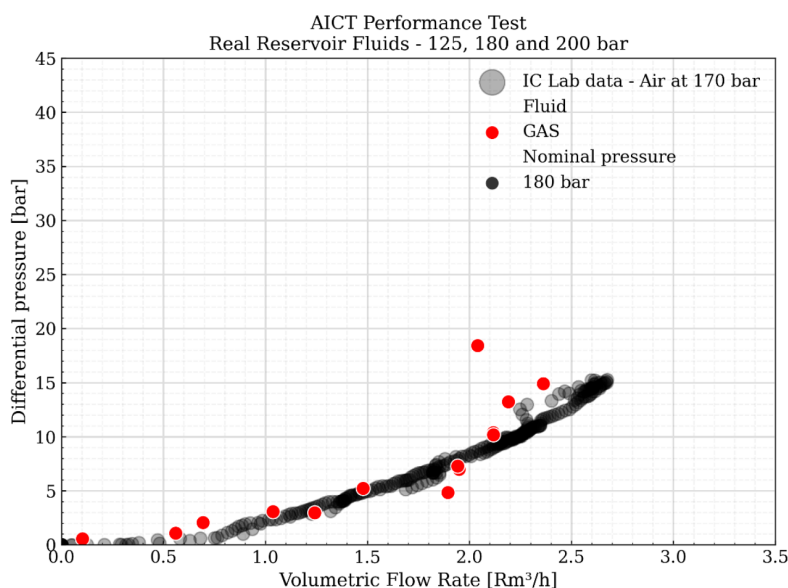


Figure 9—Performance plot with model fluid and real fluid.

Comparison of valve performance using model fluids and real fluids at equivalent densities demonstrates strong correlation, validating the use of model fluids in early-stage development. Corresponding fluid density data is shown in Fig. 10.

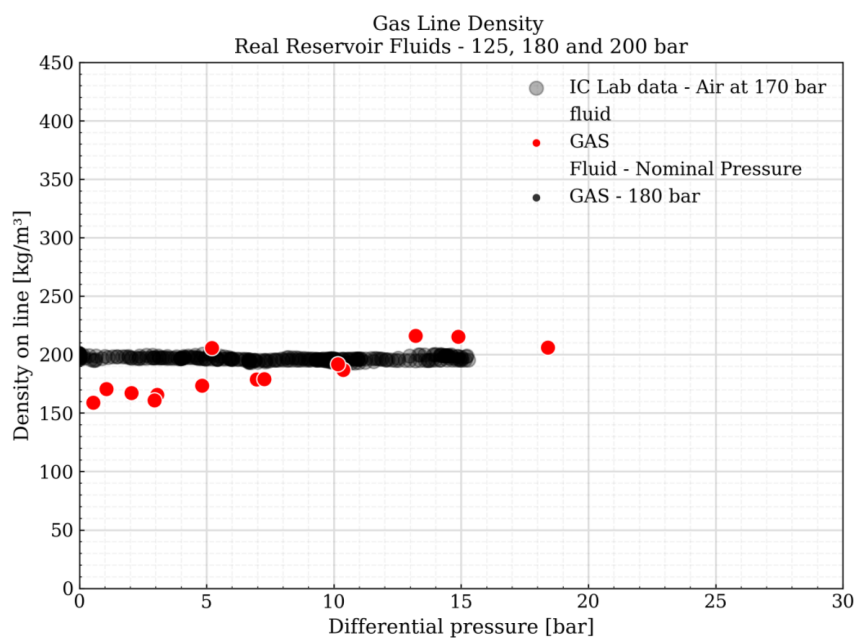


Figure 10—Measured density of model fluid and real fluid at different differential pressures. Density of the model fluid (black dots) is easier kept constant over the range of differential pressures tested.

Fig. 11 compares the water capacity when tested with fresh water at 90°C and formation water at 100°C. Some differences are seen above 15 bar.

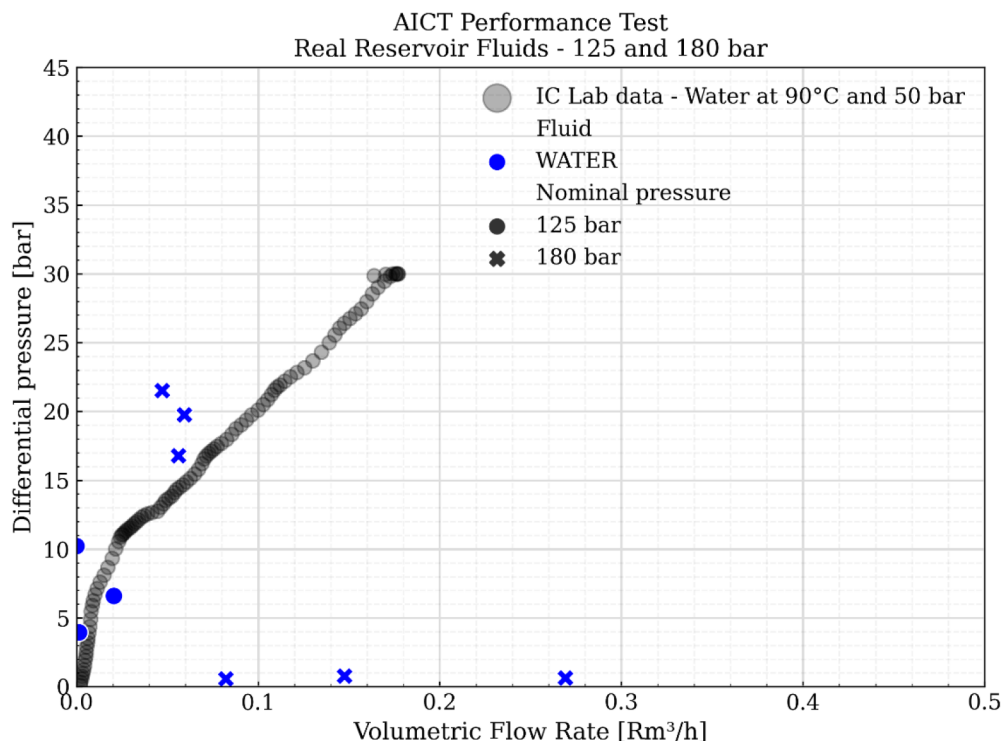


Figure 11—Performance comparison between fresh water (black) and formation water (blue) at 90 and 100°C respectively.

As discussed earlier, the valve is tuneable for a specific well condition. This specific valve was tuned to choke water up to 20 bar differential pressure, which is inside the operating region for the test rig using fresh water to avoid cavitation. The cavitation limit was established by internal testing on similar valve technologies and were performed according to API 19ICD, Annex D, section 2.42 (API SPEC 19ICD, 2020). Based on these tests the cavitation inception is found to be when the inlet pressure to the valve is equal to 2-2.5 times the differential pressure over the valve with water at 20°C. For water at higher temperatures, the inception of cavitation is expected to start earlier, due to the higher vapour pressure.

The inception of boiling in the main line would eventually choke the main flow, while boiling in pilot line could potentially reduce the pressure drop in the laminar flow element and thus lower the pressure under the piston, causing the valve to open more for water. There is no evidence of boiling and consequently choking in the main line from these tests, but the linear increase in water at around 15 bar could possibly be due to initial boiling in the pilot line, which was not seen in the reservoir condition flow loop at the same differential pressure. Accordingly, the data obtained from laboratory testing with fresh water at 90°C and pressures exceeding approximately 15 bar are subject to uncertainty, warranting further investigation. In a well environment where the annulus pressure is significantly higher, this will not cause any issue within the current design limits of the valve.

The variations observed can also be partly due to differences of the water properties and measurement uncertainties due to rates at the lower range of the Coriolis flow meter in the external flow loop. Water rate for the valve in closed position shows to be key in the below field case and this item will be a key part of future work.

Mathematical model of valve and effective zonal performance

The laboratory measurements detailed above demonstrate that the gas and condensate performance of the gas AICT varies with pressure, which is linked to changes in fluid properties. In a gas field under depletion the valves will experience a pressure drop, from approximately 330 bar to around 70 bar in the case study below. The experimental data indicates that the flow performance follows that of an orifice of a given

dimension for 100% gas and 100% water. However, the flow performance for condensate is more complex. The key focus here is the gas and water performance of the valve. For this early modelling, we simplify by assuming that the performance measured at 180 bar can serve as an average performance for all background pressures. First, a mathematical model is fitted to this performance. Next, we quantify how several valves perform together in a packer zone, including the effect of annulus phase segregation (Brekke et al., 2025; Langaas et al., 2024; Langaas et al., 2025).

Valve model

A trinary Inflow Control Device (ICD) model, specifically of the orifice type, was found to be appropriate for matching the valve performance in single-phase flow. An effective ICD diameter of 2.7 mm corresponded well with the gas performance. To match the performance for single-phase condensate (oil) and single-phase water, a phase-specific flow area fraction (f_{Ap}) was utilized. Consequently, the theoretical pressure drop of the valve in single-phase flow becomes

$$\Delta p = \frac{1}{2} \rho \left(\frac{q}{A^* f_{Ap}} \right)^2, \quad (1)$$

with f_{Ap} equal 1, 0.3 and 0.05 for gas, oil and water. This mathematical model aligns with the valve construction principle, which features a smaller flow area (pilot flow only) for water and the full flow area (pilot + main flow) for gas. The effective flow area for condensate varies across the tested pressures and corresponds to a medium fractional flow area at a back pressure of 180 bar. The water flow rates were minimal, at or below the lower boundary of the Coriolis mass flow meter for water. Consequently, the water results were cross-checked with model lab results, showing some uncertainty as discussed above. The following case study will demonstrate the critical importance of water capacity. The measured gas density varies with flow rate, showing higher density with increasing rate. The measured densities of gas, oil, and water are used in the model match shown in Fig. 12.

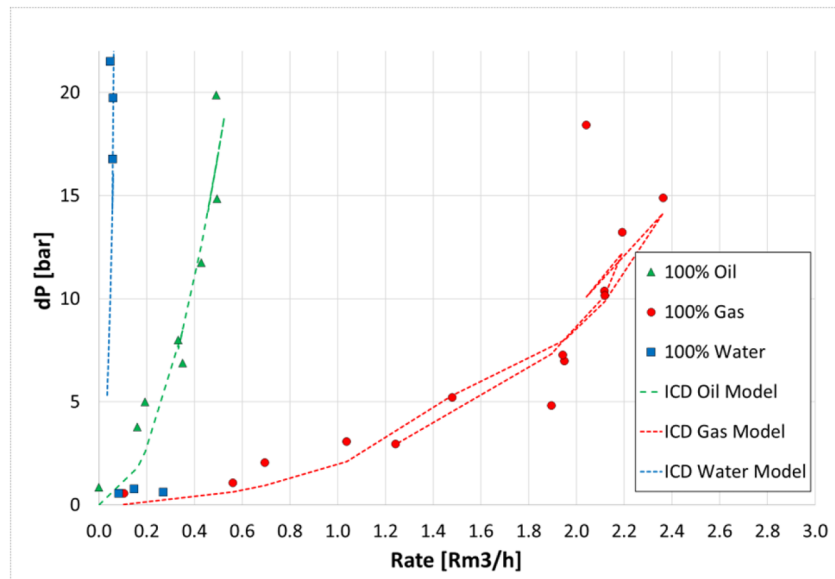


Figure 12—Trinary ICD function (stippled lines) vs. experimental data (points) for the gas AICT.

Effective zonal performance of many valves

Phase segregation happens in the annulus (outside the sand screens) if this annulus stays open and the annular axial velocities are below 0.3 m/s, as explained by Brekke et al. (2025). This valves capacity is around 2.5 Rm3/h for gas. With an annulus dimension based on outer and inner diameter of 8.5 and 5.5

inches respectively, assuming all flow to be axial, we derive axial average velocity of 0.03 m/s. Such low velocities will clearly give a stratified flow regime and annular phase segregation in an open annulus. A dedicated CFD based lower completion modelling tool has been developed to understand how several inflow control technologies work together and further derive effective zonal performance models needed in reservoir modelling. We assume further 100-meter packer spacing, as we know packer spacing will be important to limit early water production in this field case. To secure sufficient initial gas capacity, we assume two valves per screen joint, and total 16 valves for a 100-meter packer zone. The well in our case will be close to horizontal, and we assume 89° section inclination as an approximation to get the expected phase segregation with the average inlet model (Brekke et al., 2025). In reality – the valves will have a point inlet with random and discrete angle – giving phase segregation also at 90° section inclination. In this software, we can specify that the valves close when the water volumetric fraction is above a given threshold. Linked to laboratory data, 7% water fraction is assumed in this study as a closing criterion. shows one example with 5% water fraction in the reservoir inflow and shows how water will accumulate in the annulus and lead to a steady state solution where only 6 valves remain in open position while 10 valves are in closed/choked position. The CFD tool is transient, and the figure also shows the transient history of two valves, where valve nr. 8 closes when the water volume fraction gets above the closing criterion. Valve nr 6 does not get above this criterion and stays open at steady state. Oil/condensate flow is not the focus of this study, but if condensate accumulates in the annulus, it will be sitting between the gas and water as shown here.

Phase segregation occurs in the annulus (outside the sand screens) if it remains open and the annular axial velocities are below 0.3 m/s, as explained by Brekke et al. (2025). The valve's capacity is approximately 2.5 Rm3/h for gas. With an annulus dimension based on outer and inner diameters of 8.5 and 5.5 inches, respectively, and assuming all flow is axial, we derive an axial average velocity of 0.03 m/s. Such low velocities will clearly result in a stratified flow regime and annular phase segregation in an open annulus. A dedicated CFD-based lower completion modelling tool has been developed to understand how several inflow control technologies work together and to derive effective zonal performance models needed in reservoir modelling. We assume a 100-meter packer spacing, as we know packer spacing is important to limit early water production in this field case. To ensure sufficient initial gas capacity, we assume two valves per screen joint, totaling 16 valves for a 100-meter packer zone. The well in our case will be nearly horizontal, and we assume an 89° section inclination as an approximation to achieve the expected phase segregation with the average inlet model (Brekke et al., 2025). In reality, the valves will have a point inlet with random and discrete angles, resulting in phase segregation even at a 90° section inclination. In this software, we can specify that the valves close when the water volumetric fraction exceeds a given threshold. Based on laboratory data, a 7% water fraction is assumed in this study as the closing criterion. Fig. 13 shows an example with a 5% water fraction in the reservoir inflow, illustrating how water will accumulate in the annulus and lead to a steady-state solution where only six valves remain open while ten valves are closed/choked. The CFD tool is transient, and the figure also shows the transient history of two valves, where valve number 8 closes when the water volume fraction exceeds the closing criterion. Valve number 6 does not exceed this criterion and remains open at steady state. Although oil/condensate flow is not the focus of this study, if condensate accumulates in the annulus, it will be situated between the gas and water as shown here.

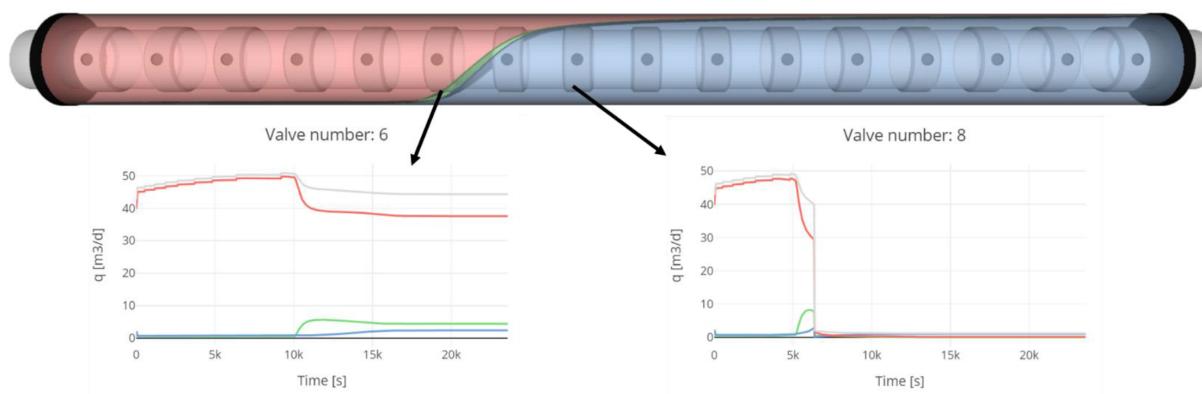


Figure 13—Lower completion modelling tool used on a 100m zone with 16 gas AICTs. Inflow into annulus is a uniform mix flow with 93% gas, 2 % oil and 5 % water. Pressure drops across the valves are around 10 bar. Upper: Final steady state solution showing how water accumulates in the annulus. (Well zone inclination is 89° while shown as perfect horizontal in this figure and not a true scale.) Lower: Transient history of valve nr. 6 and 8. Gas = red, oil = green, water = blue, total rate = grey.

Fig. 14 illustrates the steady-state conditions for all valves. The volumetric flow rate and pressure drop for each valve are consistent with the input function described earlier.

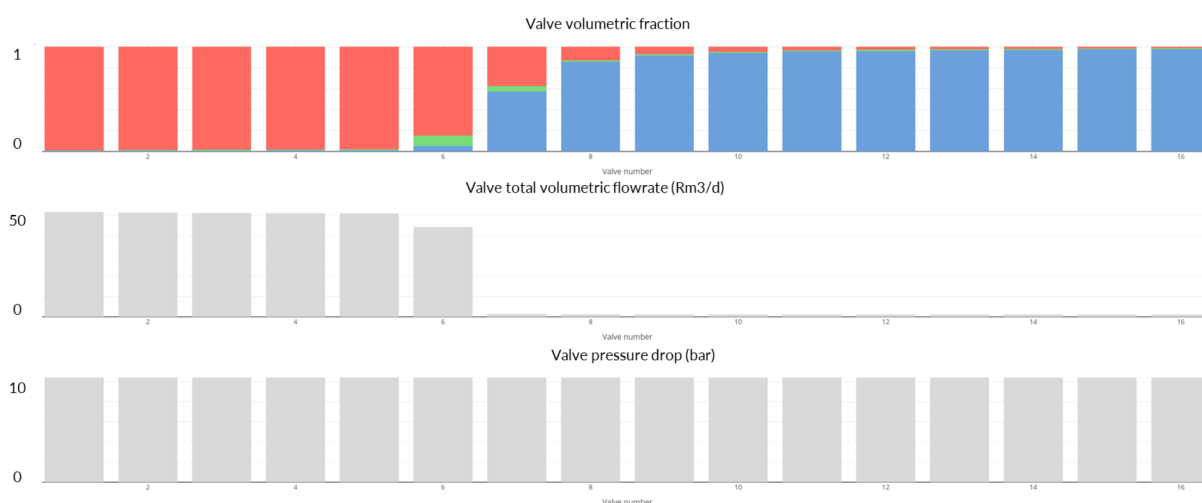


Figure 14—Lower completion modelling tool used on a 100m zone with 16 valves showing the steady state solution. Inflow into annulus is a uniform mix flow with 93% gas, 2 % oil and 5 % water. Pressure drops across valves is around 10 bar.

With a 5% water volume fraction, six valves remain open. At a 10% water volume fraction, three valves remain open. In scenarios with a 20% water volume fraction, one valve will fluctuate between open and closed positions at steady state

In some fields, the annulus will be closed. The annulus is sometimes filled with gravel to protect against erosion or becomes filled with natural sand after a sand collapse. Consequently, any mixed flow from the reservoir will enter the valve as mixed flow during steady state. If a mix with 5% water enters the zone, all valves will, in principle, detect this 5% water and remain open (assuming a closing criterion slightly higher at 7% water volume fraction). At 7% water and above, all valves will close. Therefore, with an open annulus, the autonomous choking of the zone will commence at lower water volumes and be more gradual compared to a closed annulus. This is a crucial detail to consider in lower completion design work. Table 4 compares how a packer section of 16 valves is expected to operate in the two annulus scenarios. Both scenarios will be investigated in the case study below.

Table 4—100 m zone with 16 gas AICTs and approximate steady state description with and without open annulus outside sand screens.

Water volume fraction (downhole. cond.)	Open annulus (and phase segregation)	Non-open annulus (and mixed flow)
5%	6 valves open	16 valves open
10%	3 valves open	0 valves open
20%	1 valve open and 1 valve fluctuating	0 valves open
30%	1 valve open	0 valves open
40%	1 valve fluctuating	0 valves open
50%	0 valves open	0 valves open

Reservoir and well model

A gas discovery offshore Norway is planned to be developed with a long horizontal well. Analogous field experiences indicate that early water production is a significant risk. Reservoir modelling studies show that a long well at the crest of the accumulation can optimize gas recovery. The expected gas accumulation and an initial well plan are shown in Fig. 15. One concept to manage water arrival is to use operated flow control valves placed on a stinger inside the lower completion. Another approach is to use autonomous inflow control technology that chokes water, such as the gas AICT. A vertical flow performance table generated for 5.5-inch production tubing is used to calculate the pressure drop between the lower completion and the subsea wellhead. The minimum wellhead pressure is assumed to be 50 bar, and the initial reservoir pressure is assumed to be 328 bar in the gas accumulation. To understand the potential impact of water shut-off technologies, we first conduct a simple screening exercise before performing more detailed modelling.

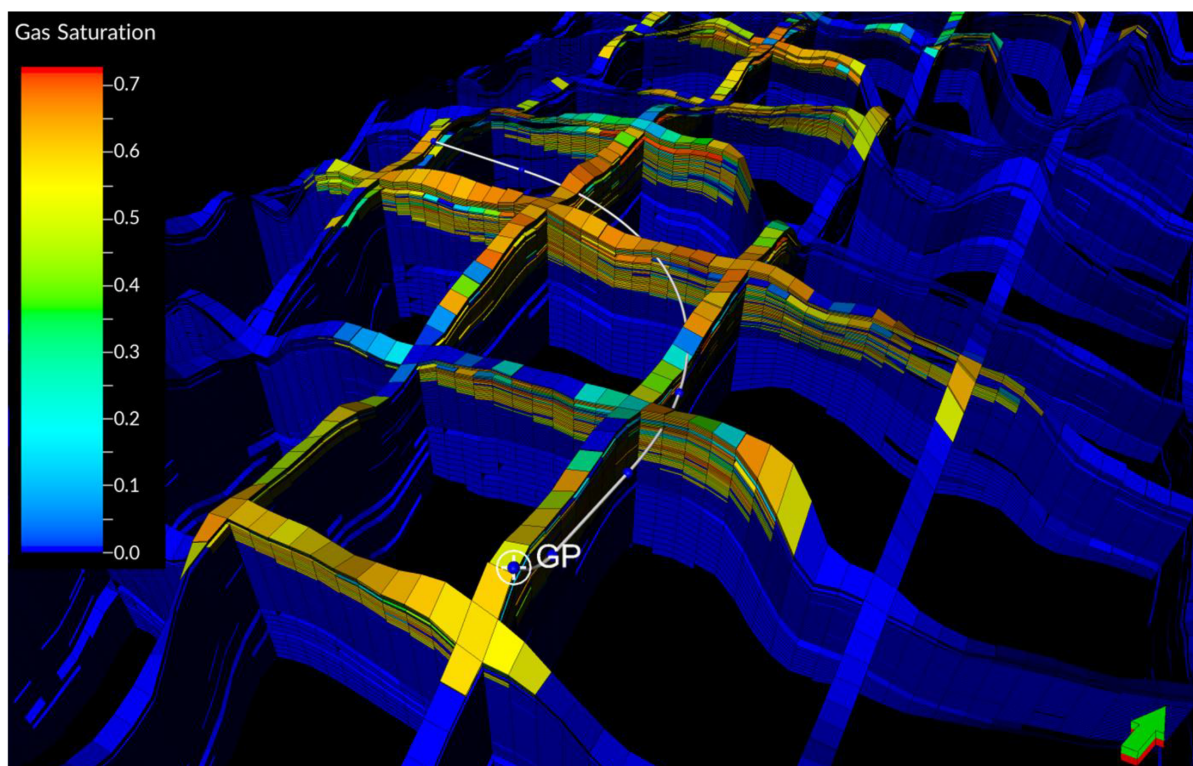


Figure 15—Dynamic model showing initial gas saturation and a planned well path. Grid size is 75*75 horizontally and around 0.5 m vertically. Well path is close to horizontal and around 2800m in the reservoir. (Model is clipped showing every 11th grid section in each horizontal direction).

Screening evaluation of water shut-off technologies

To understand the potential impact of water shut-off technologies, a fast-screening exercise was conducted. A simple well model was used, which did not include well segmentation details and therefore did not account for frictional pressure drops in the lower completion. An economic criterion was applied at the grid-well connection level, where grid connections with a certain level of water production were closed. The water-gas ratio of the grid connection was used for this purpose. Laboratory results at 180 bar indicated that the valve closed at around 7% water volume fraction, translating to approximately 0.0004 Sm³/Sm³ in water-gas ratio. Fig. 16 shows that such a closing criterion significantly improves gas recovery. In the "Base" case (no water shut off), water production increased to 1800 Sm³/d before the well ceased production due to lifting problems. In reality, the well would likely cease production earlier due to water loading during longer shut-ins. The projected profile assumes steady-state flow, which can be maintained with artificial lift systems such as gas lift, although this is not commonly used in gas wells. With water shut-off, water production is minimized, and the well produces 23% more gas volume with significantly less water. Sensitivity analyses with water-gas ratio closing criteria three times lower or higher yielded similar gas recovery results. An additional sensitivity analysis with a closing criterion of WGR=0.0028 (close to 50% water volume fraction under laboratory conditions) resulted in the same gas recovery and a maximum water production of 150 Sm³/d. This indicates that the gas AICT closing criterion has a limited impact on the field case results.

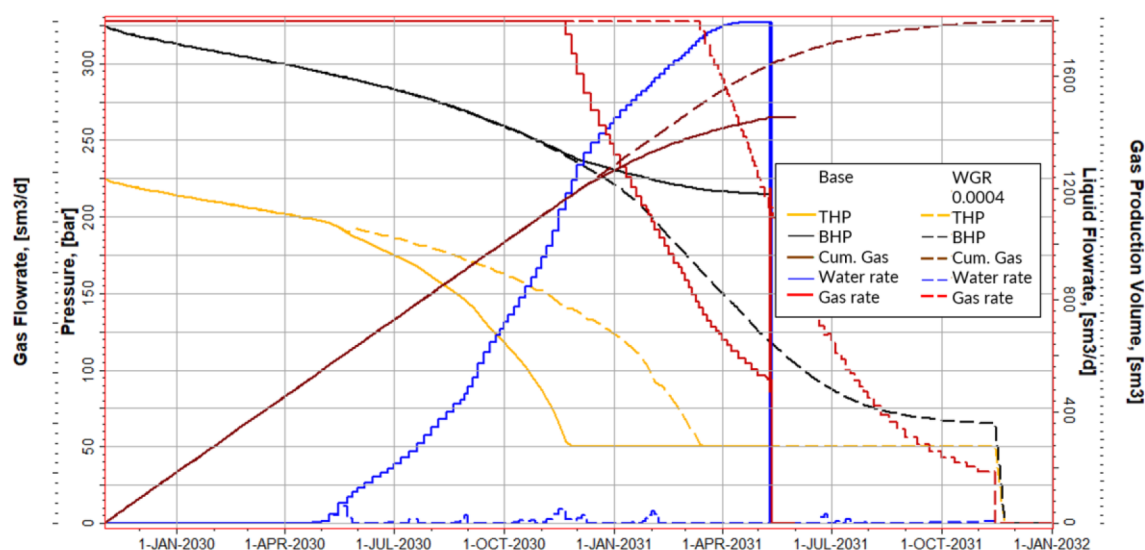


Figure 16—Simple well modelling and using water gas ratio (WGR=0.0004 Sm³/Sm³) as criterion to close grid connections. Base case has no water shut-off action. (Well gas rate hidden, same scale in all plots).

This initial estimate provides a preliminary indication of how a well equipped with the gas AICT might perform. It does not account for the additional pressure drop across the valves or the water production through closed valves, both of which would reduce the estimate. Water accumulation during extended shut-ins can also affect a well with autonomous technology, and a permanent artificial lift system (e.g., gas lift) can mitigate the risk of start-up issues. In summary, the simple screening exercise suggests significant potential benefits for water shut-off technologies in this case, although more advanced modelling is necessary for more accurate quantification.

Detailed zone and well modelling

The modern industry-standard reservoir simulator used in this study features a multi-segmented well model with options for controllable annular flow control valves (AFCVs) that can replicate zonal performance as described in Table 4. This provides a direct modelling approach that is well-suited for the gas AICT. We continue to assume 100-meter packer intervals, with each interval containing 16 gas AICTs to ensure

sufficient gas capacity. Equation 1 demonstrates that flow rate scales linearly with flow area. To model 16 gas AICTs, we use one model AFCV with a flow area 16 times that of the open gas AICT. For mixed flow (without phase segregation effects), the model valve (AFCV) transitions from open to closed/choked in one step when the water fraction at the valve exceeds 0.07 (the model AFCV flow area fraction changes from 1 to 0.05 in one step). In the case of annulus phase segregation, we can approximate the gradual choking in five steps based on Table 4, as detailed in Table 5.

Table 5—100 m zone with 16 gas AICTs assuming gas/water system and annulus phase segregation as above (downhole conditions)

Water volume fraction	Water Gas Ratio	Open valves	Closed valves	Flow area fraction
0	0	18	0	1
0.05	0.053	6	12	0.367
0.10	0.111	3	15	0.208
0.2	0.250	1.5	16.5	0.129
0.4	0.667	0.5	17.5	0.076
0.5	1.000	0	18	0.05

In this model, the well is equipped with 29 model AFCVs, each representing a 100-meter packer zone with 16 gas AICTs. For simplicity, the model is uniform; however, a more realistic design would use fewer sand screens with AICTs in zones with low productivity to reduce costs. Without any actions or choking, the same model represents a base case lower completion with a uniform ICD distribution of the same flow area as the open gas AICT (i.e., 16×2.7 mm ICDs for the 100-meter zone).

Fig. 17 details the model AFCV in zone 4, located approximately 300-400 meters from the top screen. In the mixed-flow assumption scenario, the effective performance involves a one-step choking mechanism that activates when the local water fraction meets or exceeds the closing criterion, assumed to be 7%. Consequently, gas and water rates immediately drop to low levels. In the scenario with annulus phase segregation, the choking process is more gradual, occurring over a period of four weeks for this zone. The effective model valves perform as intended.

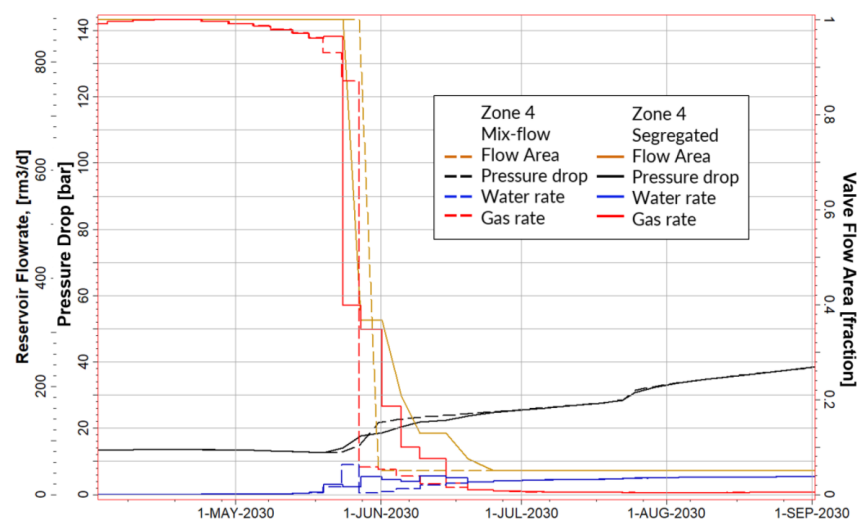


Figure 17—Model valve in zone 4 (representing a 100m zone of 16 gas AICTs) that see the first water production. For mix-flow assumption, the valve chokes to 0.05 fractional flow area when water fraction goes above 7%. When including annulus phase segregation effects, the effective choke performance is more gradual, approximated here in discrete steps. Gas and water rate in local conditions.

Fig. 18 illustrates the gas AICT well performance with and without the effects of annulus phase segregation. The more gradual choking due to annulus phase segregation results in periods of slightly higher water production. However, in the broader context, the effects are minimal, and the two model scenarios yield very similar production forecasts. Therefore, we will proceed with the mixed-flow assumption case and the one-step choking model.

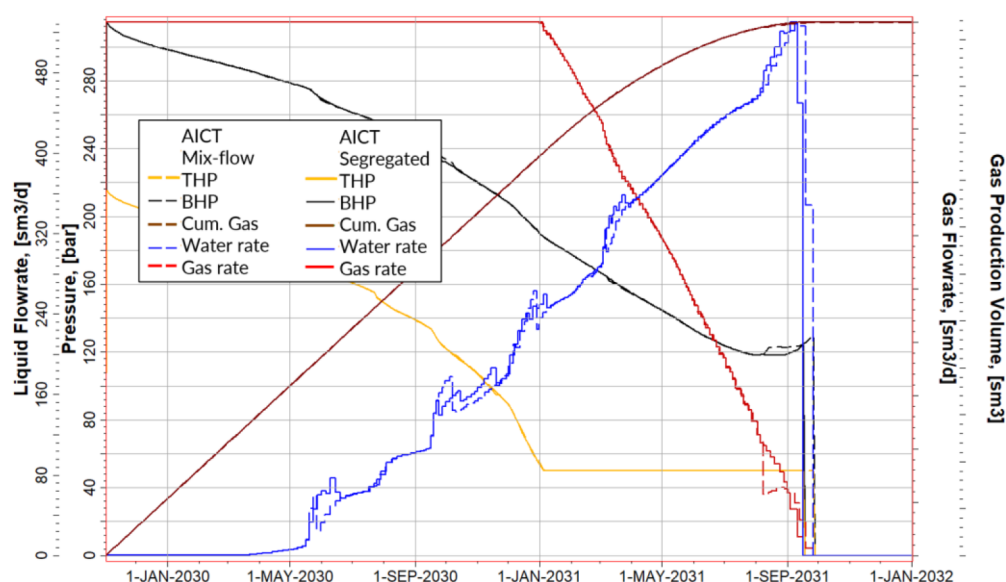


Figure 18—Well performance with gas AICTs. With and without annulus phase segregation effects. (Well gas rate hidden, same scale in all plots).

Fig. 19 compares the performance of the gas AICT completion with an ICD completion of the same flow capacity (16×2.7 mm diameter ICDs for each 100-meter packer zone). The designs are uniform, meaning the same ICD strength is applied in all zones. As expected, the gas AICT and ICD provide the same bottom hole pressure before the onset of water. The gas AICTs react when water breakthrough occurs, increasing the pressure drop of the lower completion. Zones with water are choked back, while zones without water are produced more intensively to compensate for the choked zones. The reduced water production results in a lighter well column, leading to a higher tubing head pressure (THP) and a longer gas plateau length as compared to the ICD case. The gas AICT enables less water production and more gas production. The figure also shows a sensitivity analysis for a gas AICT with a reduced flow area fraction of 0.02 in the closed position. If technically feasible, this has a positive effect. As discussed above (Fig. 11), the tested version of the valve has some uncertainty of closed flow area fraction at differential pressure above 15 bar. If larger flow area fraction than 0.05, this will have a negative impact in this case study. See also discussion on further work below.

Comparing this detailed modelling with the simplified modelling results shown in Fig. 16, we observe the following: The gas recovery in the ICD case is reduced compared to the no-action case ("Base"), as pressure drops in the lower completion are now included. The same applies to the gas AICT case. Additionally, the gas AICT produces some water in this scenario, which affects the well column weight and the tubing head pressure. The gas AICT completion produces approximately 13% more gas by January 2032 than the ICD completion, along with reduced water production. A stronger shut-off of the water zones could further increase gas recovery.

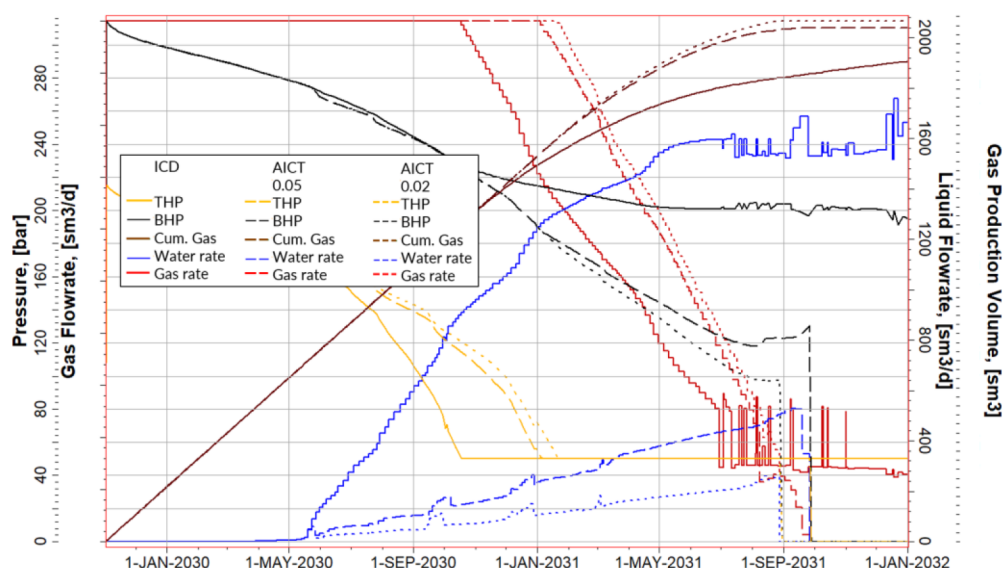


Figure 19—Well performance comparing lower completion designs with ICDs vs. gas AICTs. Mix-flow assumed. Two versions of the gas AICT model are shown, with closed flow area fraction of 0.05 and 0.02. (0.05 are closest to the experimental test data). Well gas rate hidden, same scale in all plots.

Gas fields connected to a subsea network and topside handling may face constraints on water production rates. It is also common to assume a minimum gas flow rate. A gas well without artificial lift, such as gas lift, might struggle to restart after a shut-in situation. Assuming a maximum water production of 300 Sm³/d, the comparison between ICD and gas AICT completions changes, as shown in Fig. 20. In this scenario, the impact of the gas AICT becomes more significant, extending the gas plateau period by six months. By July 2031, gas recovery is 37% higher with the gas AICT well compared to the ICD well.

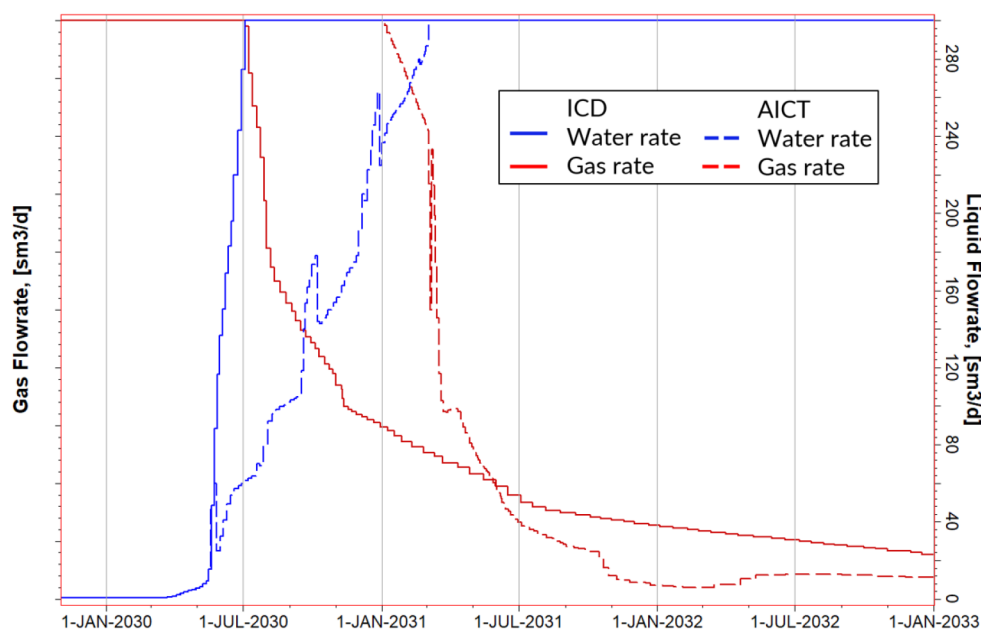


Figure 20—Well performance comparing lower completion designs with ICDs vs. gas AICTs. Mix-flow assumption and gas AICT closed flow area fraction = 0.05. Case with constrained maximum water production of 300 Sm³/d. (Well gas rate hidden, same scale in all plots).

In the modelling conducted, we did not impose any constraints on the pressure drops across the model AFCV, which represent the pressure drop across the physical gas AICTs. The pressure drop will depend on the pressure communication in the reservoir model. Three cases are shown in Fig. 21. The pink case has no

constraint on water production and a gas AICT closed flow area fraction of 0.05. Before water breakthrough, the most productive zone has a pressure drop of around 14 bar. This indicates that 16 gas AICTs (dual AICT per screen joint) are necessary for the most productive zones to ensure sufficient gas capacity. After water breakthrough and at peak choke level, seven of the zones have a pressure drop in the range of 110-140 bar. If the gas AICT closed flow area fraction is reduced to 0.02 (green case), this increases to 145-165 bar. For the violet case, with a constraint on water production and a gas AICT closed flow area fraction of 0.05, the pressure drop peak is around 100 bar. Therefore, for this field case, we ideally need AICT and annulus barriers that can tolerate high pressure differentials. Additionally, reducing the gas AICT closed flow area fraction can improve gas recovery but comes with the side effect of higher pressure drops across the AICT in the closed position. This information is useful when discussing the functional requirements of AICT for gas fields with a risk of water production.

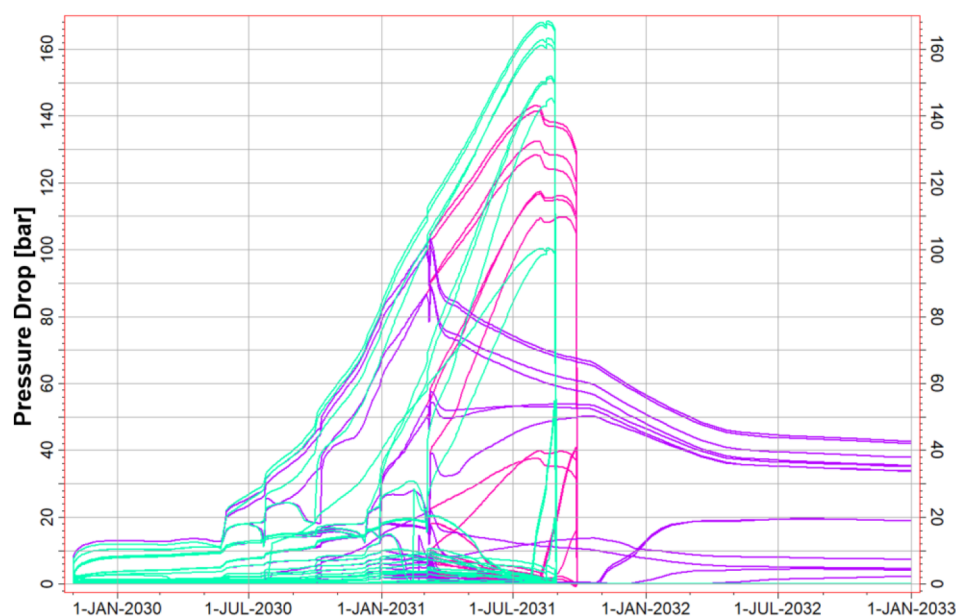


Figure 21—Model AFCV pressure drops vs time assuming mix-flow: Pink – no water production constraint, gas AICT with closed flow area fraction 0.05. Green - no water production constraint, gas AICT with closed flow area fraction 0.02. Violet – water production constraint = 300 Sm³/d, gas AICT with closed flow area fraction 0.05.

Discussion and further work

The valve demonstrates excellent performance in choking back water while remaining partially open for condensate. The proven gas capacity of the valve is sufficient for this long horizontal gas well study, assuming dual valves per screen joint in the productive zones. For shorter wells, a higher gas flow rate per valve might be necessary. Generally, a larger gas capacity than currently proven would be beneficial for future developments. The case study indicates that an ideal future gas AICT should also have a low flow area fraction in the closed position to effectively hold back water. Additionally, it must be capable of handling significant pressure differences in closed position, as demonstrated in the example provided. The choking assumptions used in the field study, i.e. constant flow area fraction of 0.05 at all differential pressure up to and above 100 bar, is outside the proved performance of the current version. Here, there are some uncertainties that will be key in future work.

Well clean-up with an autonomous valve that restricts water flow can be time-consuming. Possible solutions include intermittently forcing the valve open during flow back or by using a supporting product to enable clean-up. Both options are technically viable and will be considered in future product development.

In wet gas fields, the actual condensate properties may be heavier than the test fluid used in this study. Investigating technical solutions to ensure condensate production will be crucial in future work. One option

is to incorporate an oil AICT in each packer zone that closes for water while remaining fully open for condensate. Additionally, this AICT should be strategically placed based on geometry and steady-state modelling, as illustrated in Fig. 13.

Conclusions

1. An Autonomous Inflow Control Technology for gas wells at risk of water production ("gas AICT") has been developed using the same fluid dynamic principles as those applied in oil AICT, but with a different system design.
2. The in-house model fluid performance testing aligns well with external flow performance testing using natural gas under downhole conditions.
3. The experimental results demonstrate that the gas AICT effectively chokes water while remaining open for gas and partially open for light condensate. The exact closing criterion is challenging to measure, and a 7% water fraction has been assumed in this study.
4. A simple screening method can be used to understand the potential of water shut-off technologies. This potential can be partially realized with gas AICT or controllable Annular Flow Control Valves (AFCVs).
5. The AFCV, with an adjustable flow area based on the water-gas ratio, provides a good model representation of the effective performance of a zone with multiple gas AICTs and is used in the reservoir model study. Non-operated AFCVs also provide a good model representation of the effective performance of a zone with multiple Inflow Control Devices (ICDs), which are used as benchmark technology.
6. The gas AICT performance has been implemented in a well and reservoir model for a Norwegian gas field development, using a planned long horizontal well.
7. Annulus phase segregation is modeled in a CFD-based tool to derive effective zonal performance. Annulus phase segregation impacts zonal behavior, resulting in earlier and more gradual choking as water production increases in the zone. However, the impact on well performance in the field case is minor. The simple screening method further shows that the gas AICT closing criterion has a limited impact on the field case studied.
8. Depending on the constraints on water rate, gas cutoff rate and the assumed closed flow area fraction (0.05) at high differential pressures, the incremental gas recovery of the gas AICT compared to ICD is estimated to be between 13% and 37% in the field case study. In addition to increased gas recovery, water production is significantly reduced.
9. The field case study provides valuable input for defining functional requirements for the future development of gas AICT and associated annulus barriers.

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Nomenclature

ρ	density of the fluid at local conditions (kg/m ³)
A	Area (m ²)
AICT	Autonomous inflow control technology
API	American Petroleum Institute

CFD	Computational fluid dynamics
Cd	Discharge coefficient (dim. less)
dP	Pressure drop (bar)
BHP	Bottomhole pressure (bar)
ICD	Inflow control device
p	Pressure (bar)
q	Flow rate (m ³ /d)
THP	Tubing head pressure (bar)

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